

Lecture 1: Probability Spaces, Distributions and Expected Value

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1 The Probability Space $(\Omega, \mathcal{F}, P)$

The material of this first part of the course has to do with the study of random phenomena. The term randomness describes experiments where individual outcomes are unpredictable, but, after many repetitions, certain patterns emerge. To model a random experiment, we need a mathematical structure that rigorously defines three things: what *can* happen, which combinations of outcomes we can measure, and how likely these combinations are. This structure is the **probability space**, usually denoted by the triplet $(\Omega, \mathcal{F}, P)$.

1.1 The Sample Space: Ω

The **sample space**, Ω , is the non-empty set of all possible outcomes of an experiment. The nature of Ω dictates the type of probability theory we use. Examples of sample spaces include:

- **Finite Sample Spaces:**

- **Experiment:** Rolling a six-sided die once.

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

- **Experiment:** Tossing a coin three times.

$$\Omega = \{(H, H, H), (H, H, T), \dots, (T, T, T)\}, \quad |\Omega| = 2^3 = 8.$$

- **Countably Infinite Sample Spaces:**

- **Experiment:** Tossing a coin until the first head appears.

$$\Omega = \{H, (T, H), (T, T, H), \dots, (T, T, \dots, T, H), \dots\}.$$

This space is infinite, but we can map its elements to the natural numbers.

- **Uncountable Sample Spaces:**

- **Experiment:** Choosing a random number from the unit interval.

$$\Omega = [0, 1].$$

- **Experiment:** Measuring the exact lifetime of a light bulb.

$$\Omega = [0, \infty).$$

These spaces are infinite and uncountable, i.e. we cannot map their elements to the natural numbers.

1.2 The Event Space $(\mathcal{F})$: The Sigma-Algebra

We want to assign probabilities to **events**, which are subsets of Ω . One might naively assume we could assign a probability to *any* subset in $\mathcal{P}(\Omega)$ (the power set of Ω). This works for finite and countably infinite spaces. However, for uncountable spaces like $\Omega = [0, 1]$, this approach leads to mathematical paradoxes.

We resolve this by restricting our attention to a well-behaved collection of subsets called a **σ -algebra**, denoted $\mathcal{F}$. This is the set of “measurable” events.

Definition 1. *A collection of subsets $\mathcal{F}$ of Ω is a **σ -algebra** if it satisfies three properties:*

1. **Contains the whole space:** $\Omega \in \mathcal{F}$.
2. **Closed under complementation:** If $A \in \mathcal{F}$, then its complement $A^c = \Omega \setminus A$ is also in $\mathcal{F}$.
3. **Closed under countable unions:** If $A_1, A_2, A_3, \dots$ is a countable sequence of events in $\mathcal{F}$, then their union $\bigcup_{i=1}^{\infty} A_i$ is also in $\mathcal{F}$.

From these axioms, it follows that $\mathcal{F}$ is also closed under countable intersections and contains the empty set $\emptyset$.

Examples of σ -algebras:

- **The Trivial σ -algebra:** $\mathcal{F} = \{\emptyset, \Omega\}$.
- **The Power Set:** For a finite set Ω , its power set, i.e. the set of all possible subsets of Ω , is a σ -algebra (the largest possible). For instance, if $\Omega = \{H, T\}$, then its power set is $\mathcal{F} = \mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$.
- **The Borel σ -algebra $\mathcal{B}(\mathbb{R})$:** This is the most important σ -algebra for uncountable spaces like $\mathbb{R}$. It is defined as the *smallest* σ -algebra containing all open intervals (a, b) on $\mathbb{R}$.

1.3 The Probability Measure: $\mathbb{P}$

The final component is the **probability measure**, a function that assigns a probability to each event in our σ -algebra.

Definition 2. A **probability measure** $\mathbb{P}$ on a measurable space $(\Omega, \mathcal{F})$ is a function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ satisfying:

1. **Normalization:** $\mathbb{P}(\Omega) = 1$.
2. **Countable Additivity (σ -additivity):** For any countable sequence $A_1, A_2, \dots$ of **pairwise disjoint** events in $\mathcal{F}$ (i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$),

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

1.4 Independence

A key concept in probability is the independence of events, which formalizes the idea that the occurrence of one event does not influence the probability of another.

Definition 3 (Independence of two events). Two events $A, B \in \mathcal{F}$ are said to be **statistically independent** if and only if:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

This concept can be extended to more than two events. It is not enough for the events to be pairwise independent; a stronger condition is required.

Definition 4 (Mutual Independence). A collection of events $\{A_i\}_{i \in I}$ is said to be **mutually independent** if for any finite sub-collection of distinct indices $i_1, i_2, \dots, i_n$ from I , we have:

$$\mathbb{P}\left(\bigcap_{j=1}^n A_{i_j}\right) = \prod_{j=1}^n \mathbb{P}(A_{i_j}).$$

Example: boys and girls A common real-world assumption is that the sex of a newborn child is independent of the sex of its siblings. Let's consider a family with two children. We assume the probability of having a boy (B) is $1/2$ and the probability of having a girl (G) is $1/2$. The sample space for the sexes of the two children is $\Omega = \{(B, B), (B, G), (G, B), (G, G)\}$. Using the assumption of independence, we can calculate the following probabilities:

$$\begin{aligned}\mathbb{P}((B, B)) &= \mathbb{P}(B) \cdot \mathbb{P}(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \\ \mathbb{P}((B, G)) &= \mathbb{P}(B) \cdot \mathbb{P}(G) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \\ \mathbb{P}((G, B)) &= \mathbb{P}(G) \cdot \mathbb{P}(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \\ \mathbb{P}((G, G)) &= \mathbb{P}(G) \cdot \mathbb{P}(G) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.\end{aligned}$$

Thus, the four basic outcomes in Ω are equally likely.

Now, let's calculate the probability of the event C : "having one boy and one girl". This event corresponds to the set of outcomes $\{(B, G), (G, B)\}$. Since these outcomes are disjoint, we have:

$$\mathbb{P}(C) = \mathbb{P}((B, G)) + \mathbb{P}((G, B)) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

2 Random Variables and Distributions

An outcome $\omega \in \Omega$ can be of any nature (for instance, heads or tails in the case of coin flipping and boy or girl in the case of a newborn child). A random variable provides the crucial link from abstract outcomes to numerical values.

Definition 5. A **random variable** X is a **measurable function** from the sample space $(\Omega, \mathcal{F}, P)$ to the real numbers, $X : \Omega \rightarrow \mathbb{R}$. The term “measurable” means that for any set B in the Borel σ -algebra $\mathcal{B}(\mathbb{R})$, its pre-image under X must be an event in our original σ -algebra $\mathcal{F}$.

$$X^{-1}(B) = \{\omega \in \Omega \mid X(\omega) \in B\} \in \mathcal{F}.$$

- **Example 1: Number of Heads.** Consider an experiment of tossing two fair coins. The sample space is $\Omega = \{(H, H), (H, T), (T, H), (T, T)\}$. Let X be the random variable representing the number of heads. The function X maps outcomes from Ω to $\mathbb{R}$ as follows:

$$X((H, H)) = 2$$

$$X((H, T)) = 1$$

$$X((T, H)) = 1$$

$$X((T, T)) = 0$$

The set of possible values for X is $\{0, 1, 2\}$. This is a discrete random variable.

- **Example 2: Sum of Two Dice.** Consider rolling two six-sided dice. The sample space Ω consists of 36 ordered pairs: $\Omega = \{(1, 1), (1, 2), \dots, (6, 6)\}$. Let Y be the random variable representing the sum of the numbers on the two dice. For an outcome $\omega = (d_1, d_2) \in \Omega$, the mapping is:

$$Y(\omega) = Y((d_1, d_2)) = d_1 + d_2.$$

The set of possible values for Y is $\{2, 3, \dots, 12\}$. This is also a discrete random variable.

- **Example 3: Random Point on a Circle.** Consider a spinner that can land on any point on a circle of circumference 1, with points labeled from 0 to 1. The sample space is the interval $\Omega = [0, 1)$. Let the random variable Z be the value of the point where the spinner lands. For any outcome $\omega \in \Omega$, the mapping is simply the identity:

$$Z(\omega) = \omega.$$

The set of possible values for Z is the interval $[0, 1)$. This is a continuous random variable.

2.1 The Distribution of a Random Variable

A random variable X *induces* a probability measure on the real line. The **distribution** of X is this induced measure.

The Cumulative Distribution Function (CDF): The **CDF**, $F_X(x)$, is the universal descriptor for any random variable's distribution.

$$F_X(x) = \mathbb{P}(X \leq x).$$

The CDF is defined for all $x \in \mathbb{R}$ and is non-decreasing, right-continuous, with $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$.

2.2 Types of Random Variables and Basic Distributions

2.2.1 Discrete Random Variables

A random variable is discrete if it can take on only a finite or countably infinite number of values. Its distribution is characterized by the **Probability Mass Function (PMF)**:

$$p_X(k) := \mathbb{P}(X = k).$$

Examples include:

- **Bernoulli Distribution** ($X \sim \text{Bernoulli}(p)$): Models a single trial with two outcomes.

$$p_X(k) = \begin{cases} p & \text{if } k = 1 \\ 1 - p & \text{if } k = 0 \end{cases}$$

- **Binomial Distribution** ($X \sim \text{Binomial}(n, p)$): Models the number of successes in n independent Bernoulli trials.

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad \text{for } k \in \{0, 1, \dots, n\}$$

2.2.2 Continuous Random Variables

A random variable is continuous if its CDF is a continuous function. Its distribution can be described by a **Probability Density Function (PDF)**

$$f_X(x) = F'_X(x),$$

i.e. a function $f_X(x)$, such that

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

Examples include:

- **Uniform Distribution** ($X \sim U(a, b)$): models complete uncertainty over a fixed interval. It represents an outcome where every value in the range $[a, b]$ is equally plausible. A standard example is a random number generator that produces values in $[a, b]$. It is also often used to model rounding errors in calculations. Its PDF depend only on a and b :

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}.$$

- **Exponential Distribution** ($X \sim \text{Exp}(\lambda)$): is fundamental for modeling the time until an event occurs in a process where events happen at a constant average rate λ and independently of the time since the last event. This is known as a Poisson process. The key property of the exponential distribution is that it is *memoryless*: the probability of an event occurring in the next interval of time is independent of how much time has already elapsed. It is commonly used to model:

- The lifetime of electronic components that do not age.
- The time until a radioactive particle decays.
- The time between arrivals of customers at a service desk.

Its PDF depends on the rate of occurrence λ :

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}.$$

- **Normal (Gaussian) Distribution** ($X \sim N(\mu, \sigma^2)$): is arguably the most important distribution in all of statistics. Its prominence stems from the **Central Limit Theorem**, which states that the sum (or average) of a large number of independent and identically distributed random variables will be approximately normally distributed, regardless of the underlying distribution. This makes it an excellent model for phenomena that arise as the sum of many small, independent effects. Examples include:

- Physical measurements that are subject to many small sources of error.
- Biological characteristics like the height or blood pressure of a population.
- Returns on financial assets over a short period.

Its PDF depends on two parameters μ and σ . μ governs where the distribution is centered, while σ governs how much the distribution is spread around μ .

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad \text{for } x \in \mathbb{R}.$$

3 Multi-dimensional Random Variables (Random Vectors)

A **random vector** is a measurable function $\mathbf{X} : \Omega \rightarrow \mathbb{R}^d$, which can be written as $\mathbf{X} = (X_1, X_2, \dots, X_d)$, where each X_i is a random variable. The behavior of a random vector is described by its **joint distribution**.

- **Joint CDF**: $F_{\mathbf{X}}(x_1, \dots, x_d) = \mathbb{P}(X_1 \leq x_1, \dots, X_d \leq x_d)$.
- **Joint PMF (Discrete)**: $p_{\mathbf{X}}(k_1, \dots, k_d) = \mathbb{P}(X_1 = k_1, \dots, X_d = k_d)$.
- **Joint PDF (Continuous)**: A function $f_{\mathbf{X}}(x_1, \dots, x_d)$ such that for any region $A \subset \mathbb{R}^d$,

$$P(\mathbf{X} \in A) = \iiint \dots \int_A f_{\mathbf{X}}(x_1, \dots, x_d) dx_1 \dots dx_d.$$

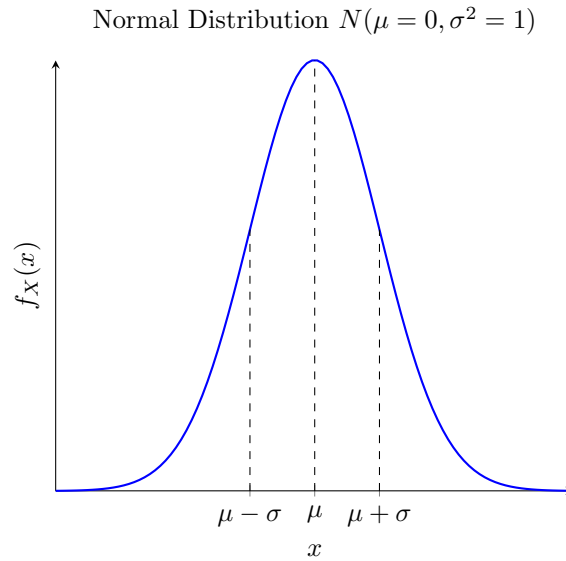


Figure 1: The probability density function of the standard normal distribution.

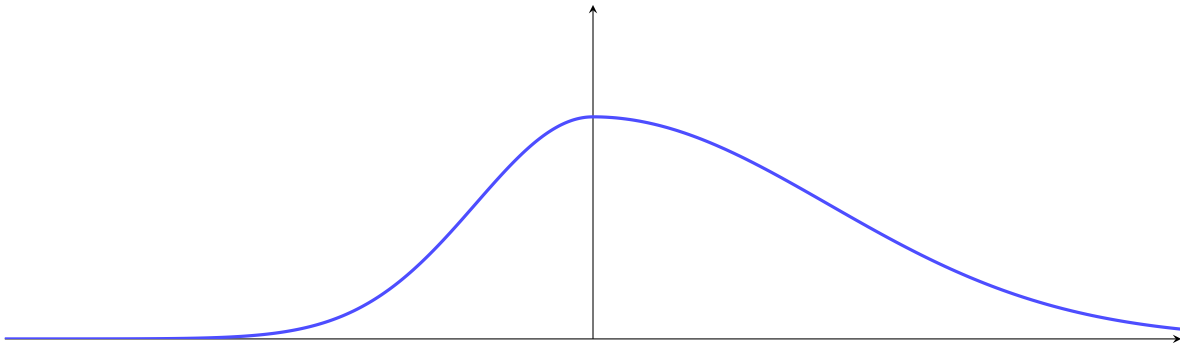


Figure 2: The probability density function of a non-normal distribution

Marginal Distributions: The distribution of a single component of a random vector is called a **marginal distribution**. For a continuous bivariate vector (X, Y) with joint PDF $f_{X,Y}(x, y)$, the marginal PDF of X is:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

3.1 Independence of Random Variables

Definition 6 (Independent Random Variables). Two random variables X and Y are **independent** if for all Borel sets $A, B \in \mathcal{B}(\mathbb{R})$:

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \cdot \mathbb{P}(Y \in B).$$

This is equivalent to their joint Cumulative Distribution Function (CDF) being the product of their marginal CDFs:

$$F_{X,Y}(x, y) = F_X(x) \cdot F_Y(y) \quad \text{for all } x, y \in \mathbb{R}.$$

For continuous variables, this implies $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$, and for discrete variables, $p_{X,Y}(k, j) = p_X(k) \cdot p_Y(j)$.

4 Expected value

The **expected value** (or expectation, or mean) of a random variable is a single number that provides a measure of its central tendency. This concept extends naturally to random vectors.

Definition 7 (expected value of a Random Variable and Vector). *Let X be a random variable.*

- If X is a **discrete** random variable with PMF $p_X(k)$, its expected value is:

$$\mathbb{E}[X] = \sum_k k \cdot p_X(k)$$

assuming the sum converges absolutely, i.e., $\sum_k |k| \cdot p_X(k) < \infty$.

- If X is a **continuous** random variable with PDF $f_X(x)$, its expected value is:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$$

assuming the integral converges absolutely, i.e., $\int_{-\infty}^{\infty} |x| \cdot f_X(x) dx < \infty$.

These are defined assuming absolute convergence.

Now, let $\mathbf{X} = (X_1, \dots, X_d)^T$ be a random vector. Its **expected value** is the vector of the expected values of its components:

$$\mathbb{E}[\mathbf{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \vdots \\ \mathbb{E}[X_d] \end{pmatrix}.$$

4.1 Functions of a Random Vector (LOTUS)

To compute the expected value of a function of a random vector, $g(\mathbf{X})$, we use the Law of the Unconscious Statistician (LOTUS), which allows us to do so without finding the distribution of $g(\mathbf{X})$ first.

Theorem 1 (LOTUS for Random Vectors). *Let $\mathbf{X}$ be a random vector in $\mathbb{R}^d$ and let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a real-valued function.*

- If $\mathbf{X}$ is discrete with joint PMF $p_{\mathbf{X}}(\mathbf{k})$: $\mathbb{E}[g(\mathbf{X})] = \sum_{\mathbf{k}} g(\mathbf{k}) \cdot p_{\mathbf{X}}(\mathbf{k})$.
- If $\mathbf{X}$ is continuous with joint PDF $f_{\mathbf{X}}(\mathbf{x})$: $\mathbb{E}[g(\mathbf{X})] = \int_{\mathbb{R}^d} g(\mathbf{x}) \cdot f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}$.

4.2 Properties of expected value for Random Vectors

The expected value operator for random vectors has several key properties. Let $\mathbf{X}, \mathbf{Y}$ be random vectors, $\mathbf{A}, \mathbf{B}$ be constant matrices, and $\mathbf{b}, \mathbf{c}$ be constant vectors.

1. **Expected value of a Constant Vector:** $\mathbb{E}[\mathbf{c}] = \mathbf{c}$.
2. **Linearity of expected value:** $\mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{B}\mathbf{Y}] = \mathbf{A}\mathbb{E}[\mathbf{X}] + \mathbf{B}\mathbb{E}[\mathbf{Y}]$. This holds even if $\mathbf{X}$ and $\mathbf{Y}$ are not independent.
3. **Product of independent random variables** If scalar random variables $X_1, \dots, X_n \in \mathbb{R}$ are independent, then $\mathbb{E}[X_1 \dots X_n] = \mathbb{E}[X_1] \dots \mathbb{E}[X_n]$.

4.3 Jensen's Inequality

Jensen's inequality relates the expected value of a convex function of a random vector to the convex function of its expected value. A function $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is **convex** if for any vectors $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$ and any $t \in [0, 1]$, $g(t\mathbf{x}_1 + (1-t)\mathbf{x}_2) \leq tg(\mathbf{x}_1) + (1-t)g(\mathbf{x}_2)$.

Proposition 2 (Jensen's Inequality). *If $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is a differentiable convex function and $\mathbf{X}$ is a random vector in $\mathbb{R}^d$ with finite expected value $\mathbb{E}[\mathbf{X}]$, then:*

$$g(\mathbb{E}[\mathbf{X}]) \leq \mathbb{E}[g(\mathbf{X})].$$

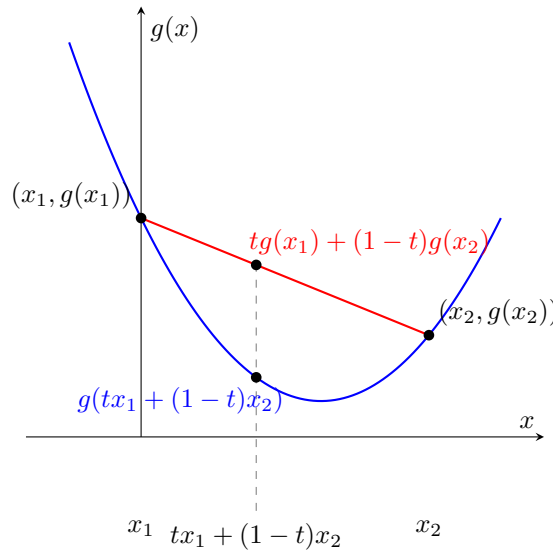


Figure 3: A convex function $f(x)$ where the secant line segment (red) connecting any two points lies above the function curve (blue). This illustrates the inequality $g(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda g(x_1) + (1 - \lambda)g(x_2)$ for any $\lambda \in [0, 1]$.

Proof. The proof relies on a key property of differentiable convex functions: they always lie on or above their tangent hyperplanes. Let $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$. The equation of the tangent hyperplane to $g(\mathbf{x})$ at the point $\boldsymbol{\mu}$ is given by $y(\mathbf{x}) = g(\boldsymbol{\mu}) + \nabla g(\boldsymbol{\mu})^T(\mathbf{x} - \boldsymbol{\mu})$, where $\nabla g(\boldsymbol{\mu})$ is the gradient of g evaluated at $\boldsymbol{\mu}$.

Because g is convex, we have that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

$$g(\mathbf{x}) \geq g(\mathbf{y}) + \nabla g(\mathbf{y})^T(\mathbf{x} - \mathbf{y}).$$

We substitute $\mathbf{X}$ for $\mathbf{x}$ and $\boldsymbol{\mu}$ for $\mathbf{y}$:

$$g(\mathbf{X}) \geq g(\boldsymbol{\mu}) + \nabla g(\boldsymbol{\mu})^T(\mathbf{X} - \boldsymbol{\mu}).$$

Now, we take the expected value of both sides. By the monotonicity of expected value:

$$\mathbb{E}[g(\mathbf{X})] \geq \mathbb{E}[g(\boldsymbol{\mu}) + \nabla g(\boldsymbol{\mu})^T(\mathbf{X} - \boldsymbol{\mu})].$$

Using the linearity of expected value on the right-hand side (and noting that $g(\boldsymbol{\mu})$ and $\nabla g(\boldsymbol{\mu})$ are constant):

$$\mathbb{E}[g(\mathbf{X})] \geq g(\boldsymbol{\mu}) + \nabla g(\boldsymbol{\mu})^T \mathbb{E}[\mathbf{X} - \boldsymbol{\mu}]$$

We know that $\mathbb{E}[\mathbf{X} - \boldsymbol{\mu}] = \mathbb{E}[\mathbf{X}] - \mathbb{E}[\boldsymbol{\mu}] = \boldsymbol{\mu} - \boldsymbol{\mu} = \mathbf{0}$. Substituting this back into the inequality gives:

$$\mathbb{E}[g(\mathbf{X})] \geq g(\boldsymbol{\mu}) + \nabla g(\boldsymbol{\mu})^T \cdot \mathbf{0}$$

$$\mathbb{E}[g(\mathbf{X})] \geq g(\boldsymbol{\mu}).$$

Substituting $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$ back gives the final result:

$$\mathbb{E}[g(\mathbf{X})] \geq g(\mathbb{E}[\mathbf{X}]).$$

This completes the proof. □

4.4 Examples of Computing expected values

Example 1: Bernoulli Distribution. Let $X \sim \text{Bernoulli}(p)$. The PMF is $p_X(1) = p$ and $p_X(0) = 1 - p$.

$$\mathbb{E}[X] = \sum_{k \in \{0,1\}} k \cdot p_X(k) = (0 \cdot (1 - p)) + (1 \cdot p) = p$$

If we perform a Bernoulli trial many times, the long-run average of the outcomes will be p . For instance, if a biased coin with a probability of heads ($X = 1$) of $p = 0.6$ is flipped 1000 times, we would expect to see about $1000 \times 0.6 = 600$ heads.

Example 2: Binomial Distribution. Let $Y \sim \text{Binomial}(n, p)$. A binomial random variable Y can be represented as the sum of n independent Bernoulli random variables: $Y = X_1 + X_2 + \dots + X_n$, where each $X_i \sim \text{Bernoulli}(p)$. Using the linearity of expected value:

$$\mathbb{E}[Y] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n p = np.$$

The expected value np is the average number of successes in n trials. If a drug has a 70% success rate ($p = 0.7$) and is given to 50 patients ($n = 50$), we would expect, on average, $50 \times 0.7 = 35$ patients to respond successfully.

Example 3: Exponential Distribution. Let $X \sim \text{Exp}(\lambda)$. The PDF is $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and 0 for $x < 0$. By the definition of expected value, we have

$$\mathbb{E}[X] = \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} x \cdot e^{-\lambda x} dx.$$

We compute this integral using integration by parts.

Let's set:

$$\begin{aligned} u &= x & dv &= e^{-\lambda x} dx \\ du &= dx & v &= -\frac{1}{\lambda} e^{-\lambda x} \end{aligned}$$

Applying the integration by parts formula:

$$\begin{aligned} \mathbb{E}[X] &= \lambda \left(\left[x \cdot \left(-\frac{1}{\lambda} e^{-\lambda x} \right) \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{\lambda} e^{-\lambda x} \right) dx \right) \\ &= \lambda \left(\left[-\frac{x}{\lambda} e^{-\lambda x} \right]_0^{\infty} + \frac{1}{\lambda} \int_0^{\infty} e^{-\lambda x} dx \right). \end{aligned}$$

Now, let's evaluate the two terms separately.

We evaluate the boundary term using limits.

$$\left[-\frac{x}{\lambda} e^{-\lambda x} \right]_0^{\infty} = \lim_{b \rightarrow \infty} \left(-\frac{b}{\lambda} e^{-\lambda b} \right) - \left(-\frac{0}{\lambda} e^0 \right).$$

The second part is clearly 0. The limit is of the indeterminate form ∞/∞ , so we can apply L'Hôpital's Rule:

$$\lim_{b \rightarrow \infty} \left(-\frac{b}{\lambda e^{\lambda b}} \right) \stackrel{L'H}{=} \lim_{b \rightarrow \infty} \left(-\frac{1}{\lambda^2 e^{\lambda b}} \right) = 0.$$

Thus, the first term evaluates to $0 - 0 = 0$.

We now evaluate the remaining integral.

$$\begin{aligned} \frac{1}{\lambda} \int_0^{\infty} e^{-\lambda x} dx &= \frac{1}{\lambda} \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_0^{\infty} \\ &= \frac{1}{\lambda} \left(\lim_{b \rightarrow \infty} \left(-\frac{1}{\lambda} e^{-\lambda b} \right) - \left(-\frac{1}{\lambda} e^0 \right) \right) \\ &= \frac{1}{\lambda} \left(0 - \left(-\frac{1}{\lambda} \right) \right) \\ &= \frac{1}{\lambda} \left(\frac{1}{\lambda} \right) = \frac{1}{\lambda^2}. \end{aligned}$$

Finally, we substitute these results back into our main equation:

$$\begin{aligned} \mathbb{E}[X] &= \lambda \left(0 + \frac{1}{\lambda^2} \right) \\ &= \lambda \cdot \frac{1}{\lambda^2} \\ &= \frac{1}{\lambda}. \end{aligned}$$

The expected value $1/\lambda$ represents the average waiting time until an event occurs. If customer arrivals at a service desk follow an exponential distribution with a rate of $\lambda = 10$ customers per hour, the average waiting time for the next customer to arrive is $\mathbb{E}[X] = 1/10$ of an hour, or 6 minutes.

Example 4: Normal Distribution. Let $X \sim N(\mu, \sigma^2)$. The PDF is $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

This integral is solved via a substitution $z = (x - \mu)/\sigma$.

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty} (\sigma z + \mu) \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\ &= \sigma \int_{-\infty}^{\infty} z \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz + \mu \int_{-\infty}^{\infty} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \\ &= \sigma \cdot 0 + \mu \cdot 1 = \mu\end{aligned}$$

The first integral is zero because its integrand is an odd function (a function h is called odd if $h(-x) = -h(x)$) over a symmetric interval. The second integral is μ times the integral of the standard normal PDF, which is 1.

The parameter μ in the definition of the normal distribution is precisely its mean. It represents the center of the symmetric distribution. For example, if adult heights are modeled by a normal distribution with $\mu = 175$ cm, then 175 cm is the average height of the population.

4.5 Markov's Inequality

The notion of expected value can be linked with the probability of our random variable taking certain values in a very general way, **independently of its general distribution**.

Proposition 3 (Markov's Inequality). *If X is a non-negative random variable ($X \geq 0$) with finite expected value $\mathbb{E}[X]$, then for any constant $a > 0$:*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. We provide the proof for a continuous random variable X with PDF $f_X(x)$. The proof for a discrete variable is analogous, with integrals replaced by sums.

By definition, the expected value of X is:

$$\mathbb{E}[X] = \int_0^{\infty} x f_X(x) dx \quad (\text{since } X \geq 0).$$

We can split the integral at the point a :

$$\mathbb{E}[X] = \int_0^a x f_X(x) dx + \int_a^{\infty} x f_X(x) dx.$$

Since $x \geq 0$ and $f_X(x) \geq 0$, the first term $\int_0^a x f_X(x) dx$ is non-negative. Therefore, we can establish the following inequality:

$$\mathbb{E}[X] \geq \int_a^{\infty} x f_X(x) dx.$$

Within the range of this second integral, we know that $x \geq a$. We can therefore replace x with a inside the integral, which may further decrease its value:

$$\mathbb{E}[X] \geq \int_a^{\infty} a f_X(x) dx$$

Since a is a constant, we can pull it out of the integral:

$$\mathbb{E}[X] \geq a \int_a^{\infty} f_X(x) dx.$$

The remaining integral, $\int_a^{\infty} f_X(x) dx$, is precisely the definition of $\mathbb{P}(X \geq a)$. Substituting this back gives:

$$\mathbb{E}[X] \geq a \cdot \mathbb{P}(X \geq a).$$

Rearranging the terms to solve for the probability gives the desired result, provided $a > 0$:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

This completes the proof. □

Bound on Wealth Inequality. Suppose the average annual income in a city is \$50,000. What is the maximum possible percentage of the population that can have an income of at least \$500,000?

Let X be the income of a randomly chosen person. We know $X \geq 0$ and $\mathbb{E}[X] = 50,000$. We want to bound $\mathbb{P}(X \geq 500,000)$. Using Markov's inequality with $a = 500,000$:

$$\mathbb{P}(X \geq 500,000) \leq \frac{\mathbb{E}[X]}{a} = \frac{50,000}{500,000} = \frac{1}{10}.$$

Thus, at most 10% of the population can have an income of \$500,000 or more. This hard limit holds true regardless of the specific distribution of income.

Performance Guarantees for Randomized Algorithms. Consider a randomized algorithm whose expected running time is 1 second. What is the maximum amount of times it will take more than 4 seconds to finish?

Let X be the algorithm's running time. We know $X \geq 0$ and $\mathbb{E}[X] = 1$ s. We want to bound $\mathbb{P}(X \geq 4)$.

$$\mathbb{P}(X \geq 4) \leq \frac{\mathbb{E}[X]}{4} = \frac{1}{4}.$$

Thus, there is at most a 25% chance that the algorithm will take 4 times its expected runtime or longer. This provides a probabilistic performance guarantee, assuring us that long delays are relatively unlikely. If we run the algorithm a sufficiently large number of times, it will take more than 4 seconds, no more than 25% of the runs.